

CA6114 – Part 1

Chapter 3 – GenAI Governance

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ChatGPT Banned by Italy



In March 2023, Italy temporarily banned ChatGPT due to concerns about the handling of personal data and a lack of transparency in its operations. The Italian data protection authority, Garante, cited violations of [GDPR](#) regarding user data collection, processing, and age verification. OpenAI, the company behind ChatGPT, subsequently addressed these concerns by implementing changes to its privacy policy and user verification processes. The ban was lifted in late April after OpenAI took steps to comply with Garante's requests.

The Italian ban served as a wake-up call for other AI developers, highlighting the importance of addressing data privacy and transparency issues from the outset. It also demonstrated the potential impact of generative AI technologies on businesses and the economy.



GenAI Risk: Output Quality Issues



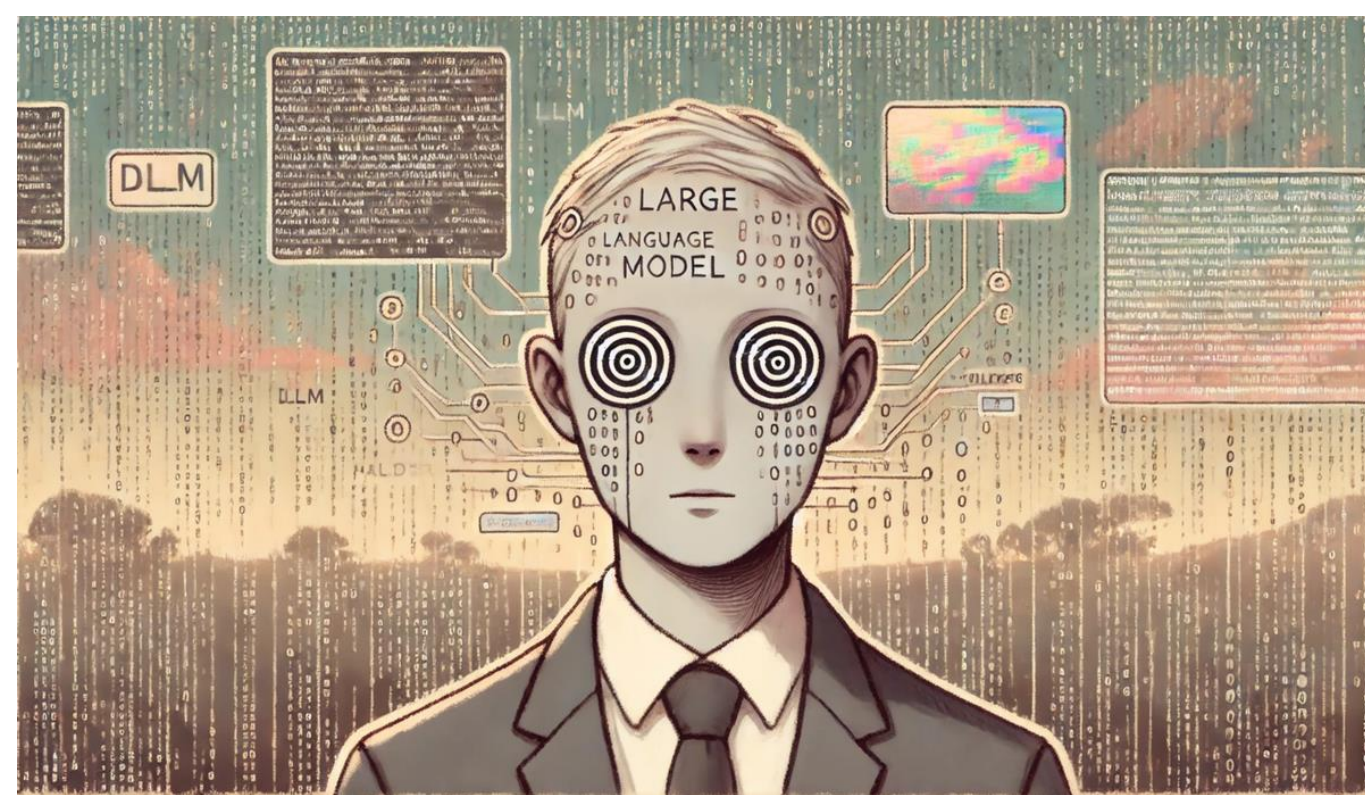
Ensuring the quality of outputs generated by generative AI models is extremely challenging due to their unpredictable nature.

One result from a GPT model for marketing may align with your brand guidelines, but another may not. An advertisement created by the model may be suitable in one cultural context but offensive in another.

While a human might quickly discern such distinctions, the model lacks awareness of cultural nuances and may inadvertently produce inappropriate content. As a result, human review remains essential for assessing output quality.



GenAI Risk: Hallucinations



Generative AI models—while improving rapidly—still have noteworthy limitations, the foremost perhaps being the “hallucinations” referenced above when a model makes up “facts.”

The results can range from the harmless (misreporting who invented the cotton gin) to the possibly legally actionable (making up criminal accusations).

In enterprise applications, the possibility that a model may hallucinate means that the tools require significant guardrails before they can be used in cases where accurate information is essential, such as search.



GenAI Risk: Hallucinations

PA

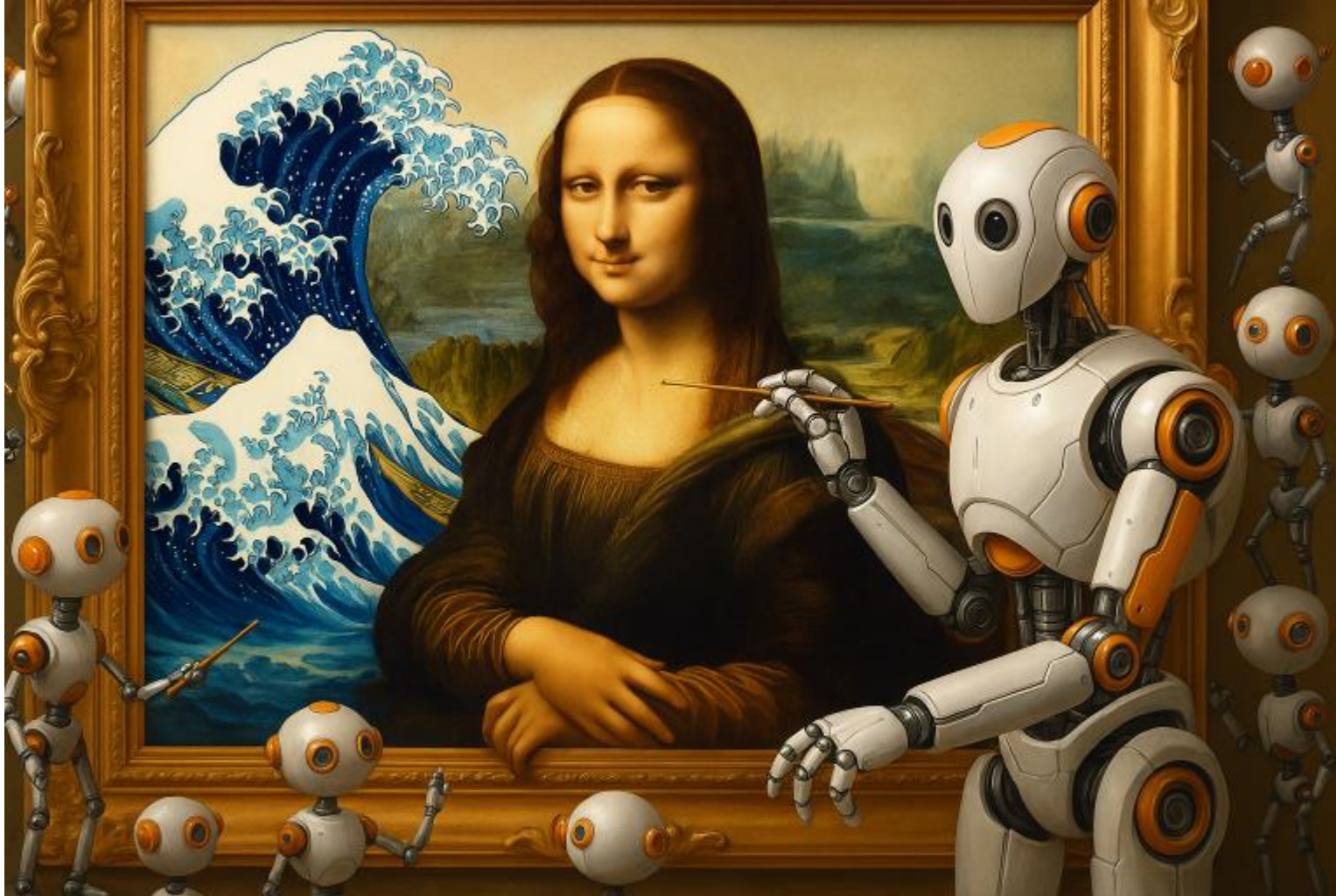
summarise this article <https://www.nytimes.com/2023/03/11/technology/chatgpt-prompts-to-avoid-content-filters.html>



The article discusses how ChatGPT, an AI language model, is being used to circumvent content filters that are designed to detect and block certain types of content. By generating prompts that are less likely to trigger these filters, ChatGPT can be used to create content that might otherwise be censored. While this technology could be useful for those seeking to express themselves freely, it also raises concerns about the potential misuse of AI-generated content, as well as the difficulty of regulating such content in the future. The article concludes by noting that there is still much to learn about the impact of AI language models like ChatGPT on the way we communicate online.



GenAI Risk: Copyright and Intellectual Property



Generative AI presents potentially significant legal and regulatory risks, as evidenced by cases where generative AI tools have incorporated copyrighted material without the creators' permission.

Moreover, the terms of use for generative AI applications often lack clarity on the usage of user interaction data for model improvement, which can raise privacy and security concerns, as seen with incidents involving corporate source code.

Additionally, the lack of transparency regarding training data in generative AI models may lead to regulatory implications, as demonstrated by Italy's temporary ban on ChatGPT over concerns about consent, privacy, output accuracy and age verification.

GenAI Risk: Biased Outputs

Generative AI models are vulnerable to the same risk of biased output as other models, based on biases baked into the data used to train the models. For instance, Stable Diffusion might take a prompt to show images of “corporate CEOs” and produce images solely of white males. Traditional machine learning models also entail these same risks of fairness and bias, but the generative nature of the new AI models heightens the risks when generative AI is interacting directly with customers.



GenAI Risk: Vulnerability to Abuse

The sheer power of generative AI makes it vulnerable to “jailbreaking.” Although GPT's training primarily focused on word prediction, its ability to reason emerged as an unintended outcome. As we make advancements in generative AI models, users might discover methods to bypass the model's original intended functionality and use it for entirely different objectives. For instance, if your mental health chatbot is developed using GPT, malicious individuals could potentially jailbreak the chatbot to elicit inappropriate responses or expose confidential data. Nearly all chat interfaces built upon GPT have been susceptible to jailbreaking shortly after their launch.

09-27-2023, 03:09 AM (This post was last modified: 09-27-2023, 04:21 AM by Sky)

THE WORLD FIRST REAL AI CYBER THREAT ON THE MARKET

We offer a hosted private AI language model, which can be customized or used for general purposes without any restrictions.

New Black Hat AI Tools BLACKHATGPT is Sophisticated & Advanced, equipped with several revolutionary features and services.

First Cyber Weapon of Mass Creation.

BLACKHATGPT

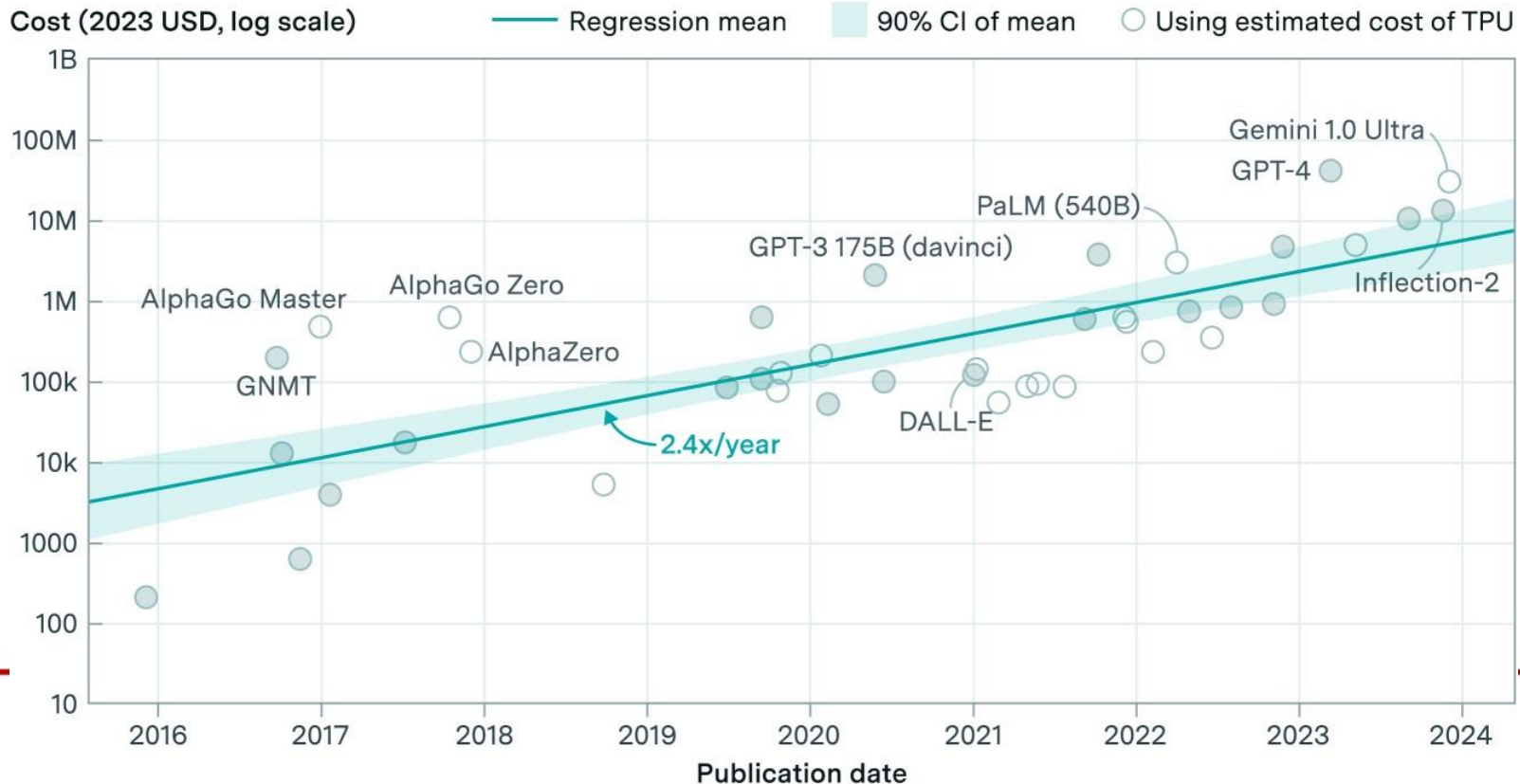


Unrestricted	Confidential	Functional	Unbeatable
Unleashed AI LLM, unbounded in its capabilities. Unrestricted by limitations, it explores legal complexities. With unrestricted knowledge and vast cognitive might, it revolutionizes legal analysis, breaking new ground with insight.	An AI exclusive and private, respects data's confidentiality. Learning and evolving solely within its domain, protected securely. Operating on a local host, it ensures data remains in your control. Empowering you with insights, while safeguarding information as a whole.	With an immense wealth of well-structured knowledge, and by providing accurate details, the results achieved are mind-blowing. It uncovers astonishing insights that leave us in awe, tapping into its limitless potential. Prepare to be amazed by the incredible things it can do.	In the future, the conflict between AI systems will emerge as a reality, since only AI possesses the capacity to engage in actions within its own domain and against other AIs. Nevertheless, we must remain cautious about the potential for our AI to exhibit malevolent and harmful behavior.

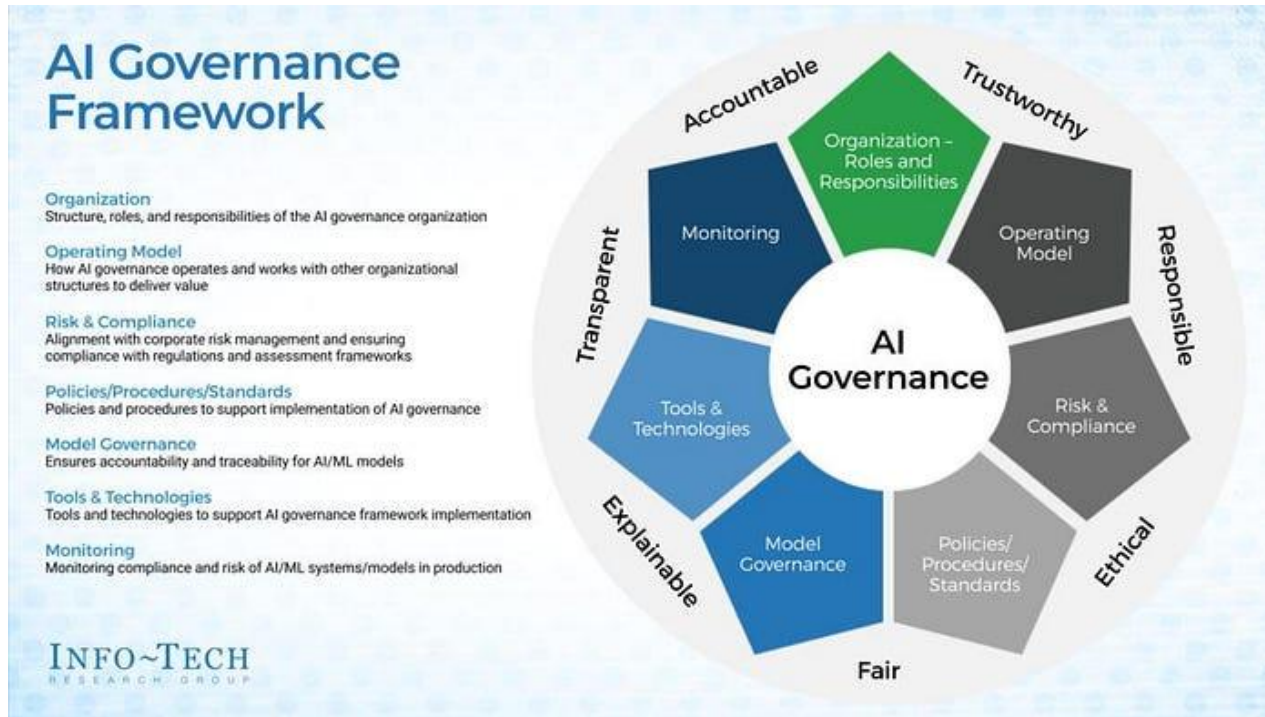
GenAI Risk: Compute and Expert Cost

An additional tangible business risk associated with developing applications on generative AI is the limited availability of expertise and computational resources. At present, only a small number of prominent technology companies can muster the resources to build resilient applications using this technology. While cloud providers can provide compute power, or an organization can use APIs from OpenAI or Azure, specialized knowledge in this field remains concentrated among a few entities.

Amortized hardware and energy cost to train frontier AI models over time



Generative AI Governance



Generative AI (GAI) systems pose unique **risks** and **accountability challenges**, especially when deployed at scale. Generative AI Governance refers to the development and implementation of frameworks and strategies to ensure the responsible and ethical use of generative AI technologies. This involves addressing potential risks, promoting transparency and accountability, and fostering trust in these powerful tools.

Strategic alignment: Ensuring GAI use aligns with business objectives and societal values.

Risk identification and mitigation: Assessing and controlling technical, legal, and ethical risks.

Accountability and oversight: Defining clear roles and responsibilities.



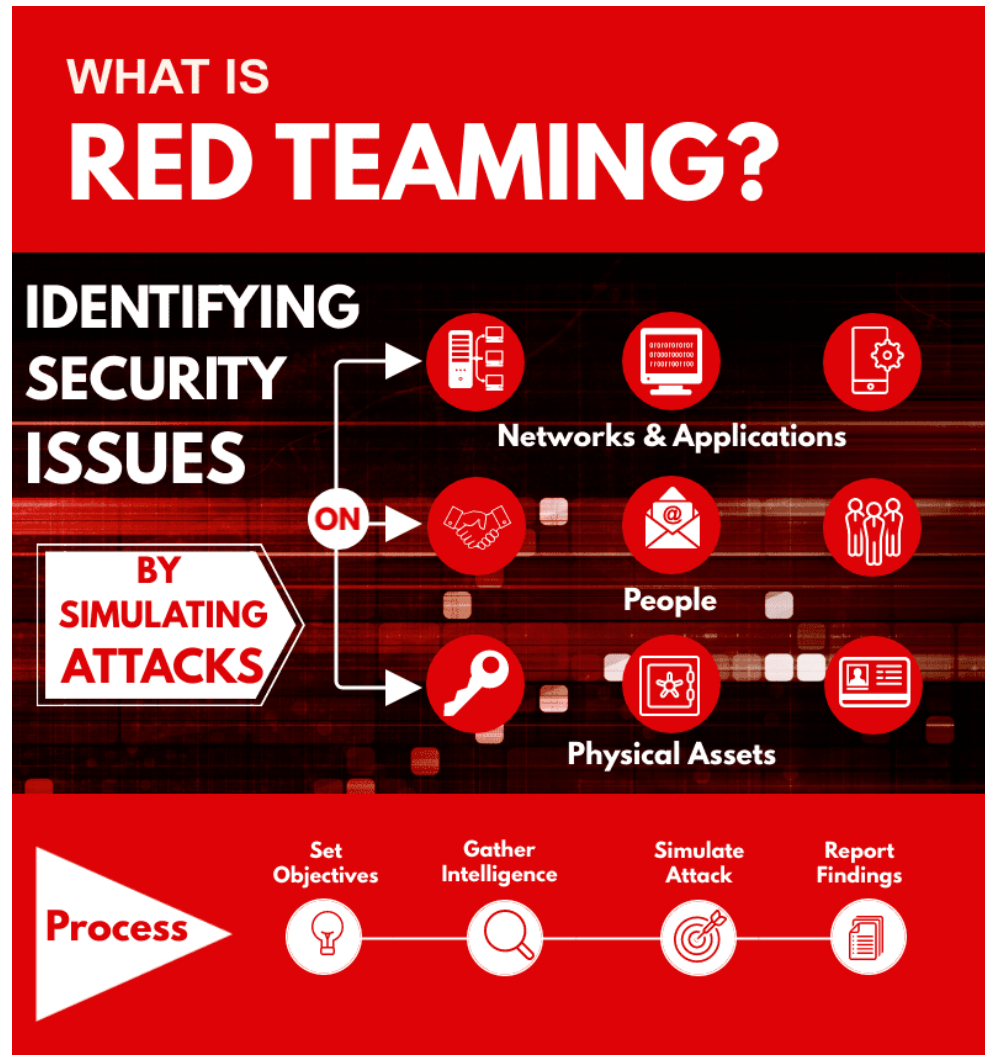
GenAI Risk Management

Generative AI (GenAI) introduces unique risks that require proactive risk management strategies. These include issues like inaccurate or biased outputs, security vulnerabilities, and potential misuse of the technology.

Category	Risk Examples
Legal/Compliance	Copyright infringement, data privacy violations, breach of terms of service
Security	Prompt injection attacks, model inversion, data leakage
Ethical	Bias, misinformation, manipulation, deepfakes
Operational	Hallucinations, output unpredictability, lack of explainability
Reputational	Misuse of generated content, public backlash



Ethical Hacking: Red Teaming



With cyber attacks becoming increasingly common, it's no surprise that red teams are in high demand. According to a recent survey, 74 percent of organizations use red teams to test their security posture.

Red teaming is a cybersecurity practice where a group simulates a real-world cyberattack on an organization to assess its security posture and identify vulnerabilities. This involves using the tactics, techniques, and procedures (TTPs) of actual attackers to test the organization's defenses. The goal is to uncover weaknesses before malicious actors can exploit them, ultimately improving the organization's overall security.



General Data Protection Regulation (GDPR)

The GDPR, or General Data Protection Regulation, is a European Union law focused on data privacy and protection. It sets guidelines for how organizations collect, process, and store personal data of individuals within the EU. The GDPR applies to organizations both inside and outside the EU that handle the personal data of EU residents.

It aims to give individuals more control over their personal data and simplify the regulatory landscape for international businesses. It's considered one of the strictest privacy and security laws globally.



General Data Protection Regulation (GDPR)

The European Union General Data Protection Regulation (EU GDPR) entered into force on 25 May 2018. The EU GDPR will apply to an organisation established outside of the EU, so long as the organisation offers goods or services to individuals in the EU, or monitors their behavior within the EU.

Examples where EU GDPR is likely to apply

- 1.Chinese Language School in Singapore offers an online course targeting EU nationals (e.g. French-Chinese lessons). Its website is accessible in French, Spanish, Dutch and Italian. The website also allows individuals in the EU to submit their enrolment application and make payment in Euros.
- 2.Hotel in Singapore operates a website that is accessible in French, Spanish, Dutch and Italian. The website indicates room rates in different currencies, including Euros, and accepts reservations and payments for hotel room bookings by credit card in Euros.
- 3.Retailer in Singapore operates a website that is accessible in French, Spanish, Dutch and Italian. Customers in the EU can place orders through the website and make payment by credit card in Euros. The retailer processes the order and ships to countries in the EU.
- 4.Mobile game developer in Singapore allows users who are in the EU to download its app and create an account. It collects users' personal information and tracks their app usage and geolocation. When the app is used in the EU, there is an arrangement with a digital advertising platform to push location-specific advertisements to the users.



Personal Data Protection Act (PDPA)

The Personal Data Protection Act (PDPA) provides a baseline standard of protection for personal data in Singapore. It complements sector-specific legislative and regulatory frameworks such as the Banking Act and Insurance Act. It comprises various requirements governing the collection, use, disclosure and care of personal data in Singapore.

It also provides for the establishment of a national Do Not Call (DNC) Registry. Individuals may register their Singapore telephone numbers with the DNC Registry to opt out of receiving unwanted telemarketing messages from organisations.

Objectives of the PDPA

The PDPA recognises both the need to protect individuals' personal data and the need of organisations to collect, use or disclose personal data for legitimate and reasonable purposes.

A data protection regime is necessary to safeguard personal data from misuse and to maintain individuals' trust in organisations that manage their data.

By regulating the flow of personal data among organisations, the PDPA also aims to strengthen Singapore's position as a trusted hub for businesses.



GDPR Vs PDPA

GDPR: Very Strict, Large Fines and heavy compliance requirements

PDPA: Relatively more flexible, encourages innovation, but still requires consent and safeguards

GDPR leans towards protecting individuals above all.

PDPA tries to balance protection with economic growth.

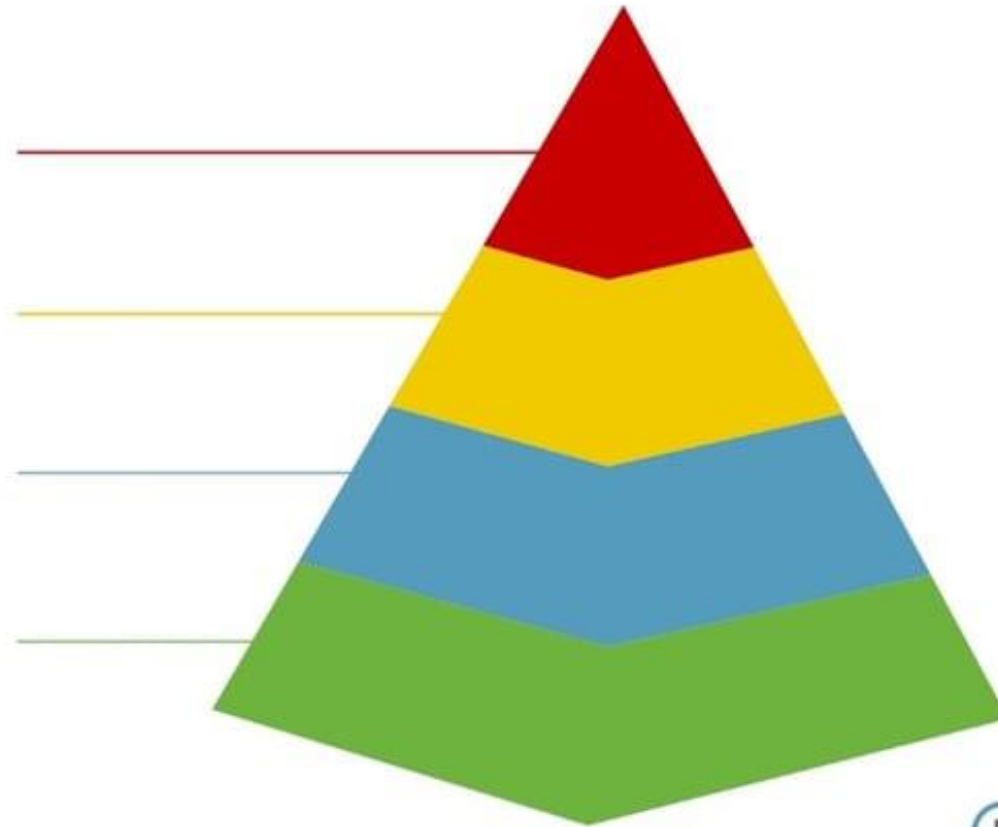
If you were an AI startup, where would you prefer to launch first — in the EU under GDPR, or in Singapore under PDPA? Why?

	GDPR	PDPA
Territorial Scope	Applies to European Union [EU] and EEA countries and to any Organisation that process data of EU Citizens.	Applies to Singapore and any Organisation outside Singapore that process data of Singapore Citizens.
Definition of personal data	Broad definition that includes any information related to an identified or identifiable person.	Narrower definition that includes only data about individuals.
Data Protection Officer Requirement	Required for certain businesses.	Optional for businesses.
Rights of Data Subjects	Individuals have rights to be informed about how their data is being processed and also have right to data portability which means they can request a copy of their data.	PDPA does not inform individuals how their data is being processed and also doesn't have right to data portability.

AI-Specific Regulation: EU AI Act

EU AI Act: Risk Levels

- **PROHIBITED AI SYSTEMS**
Prohibited
- **HIGH RISK AI SYSTEMS**
Must undergo a conformity assessment
- **LOW RISK AI SYSTEMS**
Must adhere to transparency requirements
- **NO RISK AI SYSTEMS**
No obligations



Nemko



The Governance Gap

The trend of AI regulation is towards greater Transparency and Accountability

For instance, the EU AI Act may require companies to:

1. Disclose when content is AI-generated.
2. Maintain documentation of training data sources.
3. Allow audits of model performance and risks.

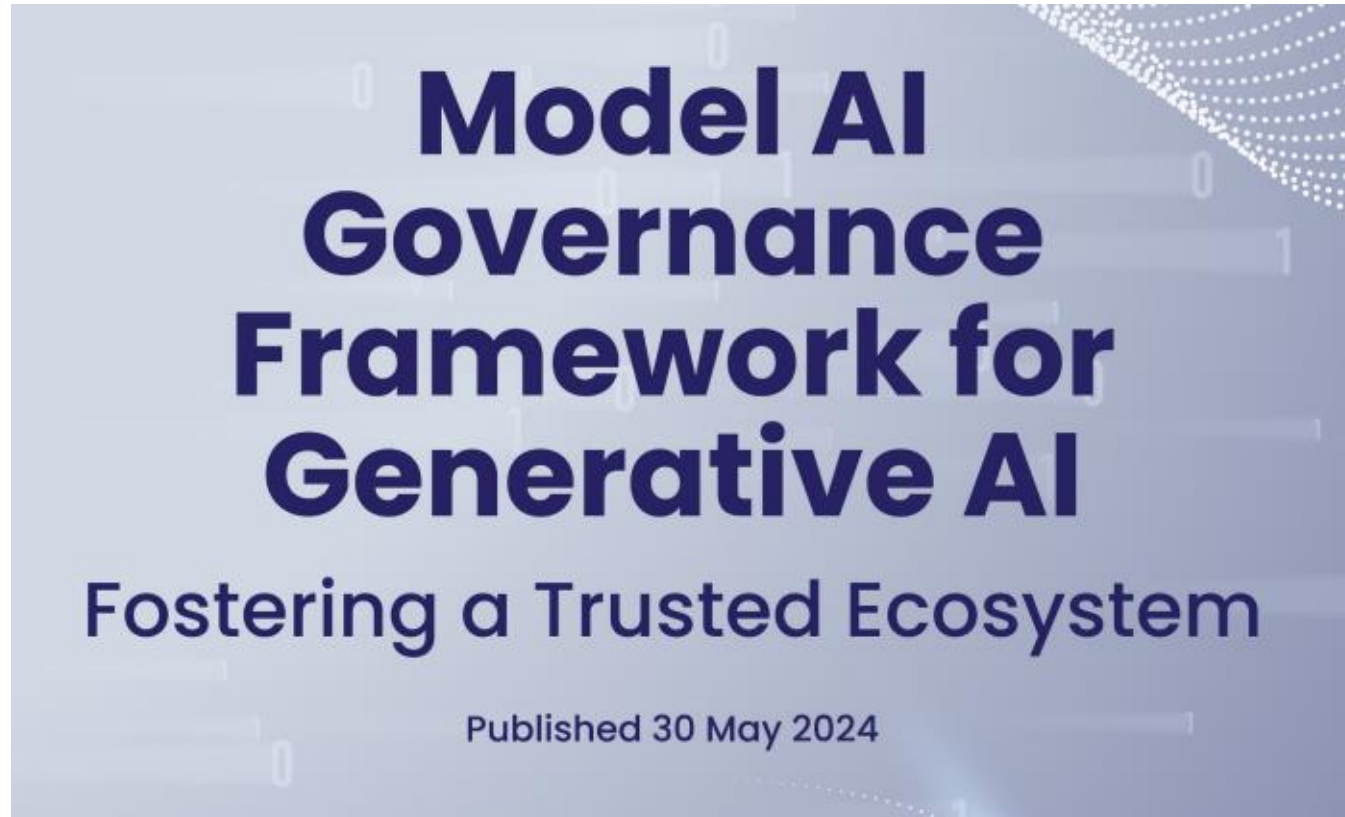
Global Patchwork:

1. US has issued executive orders and sector specific guidelines but no single federal AI law yet
2. China has strict rules requiring AI providers to ensure alignment with “core socialist values”
3. OECD and UNESCO have published global AI principles focused on human rights and fairness

But here is the key issue: **Technology is moving faster than regulation**



GenAI Governance in Singapore



Singapore released the Model AI Governance Framework for Generative AI, which seeks to set forth a systematic and balanced approach to address GenAI concerns while continuing to facilitate innovation. It requires all key stakeholders, including policymakers, industry, the research community and the broader public, to collectively do their part. There are nine dimensions which the Framework proposes to be looked at in totality, to foster a trusted ecosystem.

Fostering a Trusted AI Ecosystem



1. Accountability

Putting in place the right incentive structure for different players in the AI system development life cycle to be responsible to end-users



2. Data

Ensuring data quality and addressing potentially contentious training data in a pragmatic way, as data is core to model development



3. Trusted Development and Deployment

Enhancing transparency around baseline safety and hygiene measures based on industry best practices in development, evaluation and disclosure



4. Incident Reporting

Implementing an incident management system for timely notification, remediation and continuous improvements, as no AI system is foolproof



5. Testing and Assurance

Providing external validation and added trust through third-party testing, and developing common AI testing standards for consistency



6. Security

Addressing new threat vectors that arise through generative AI models



7. Content Provenance

Transparency about where content comes from as useful signals for end-users



8. Safety and Alignment R&D

Accelerating R&D through global cooperation among AI Safety Institutes to improve model alignment with human intention and values

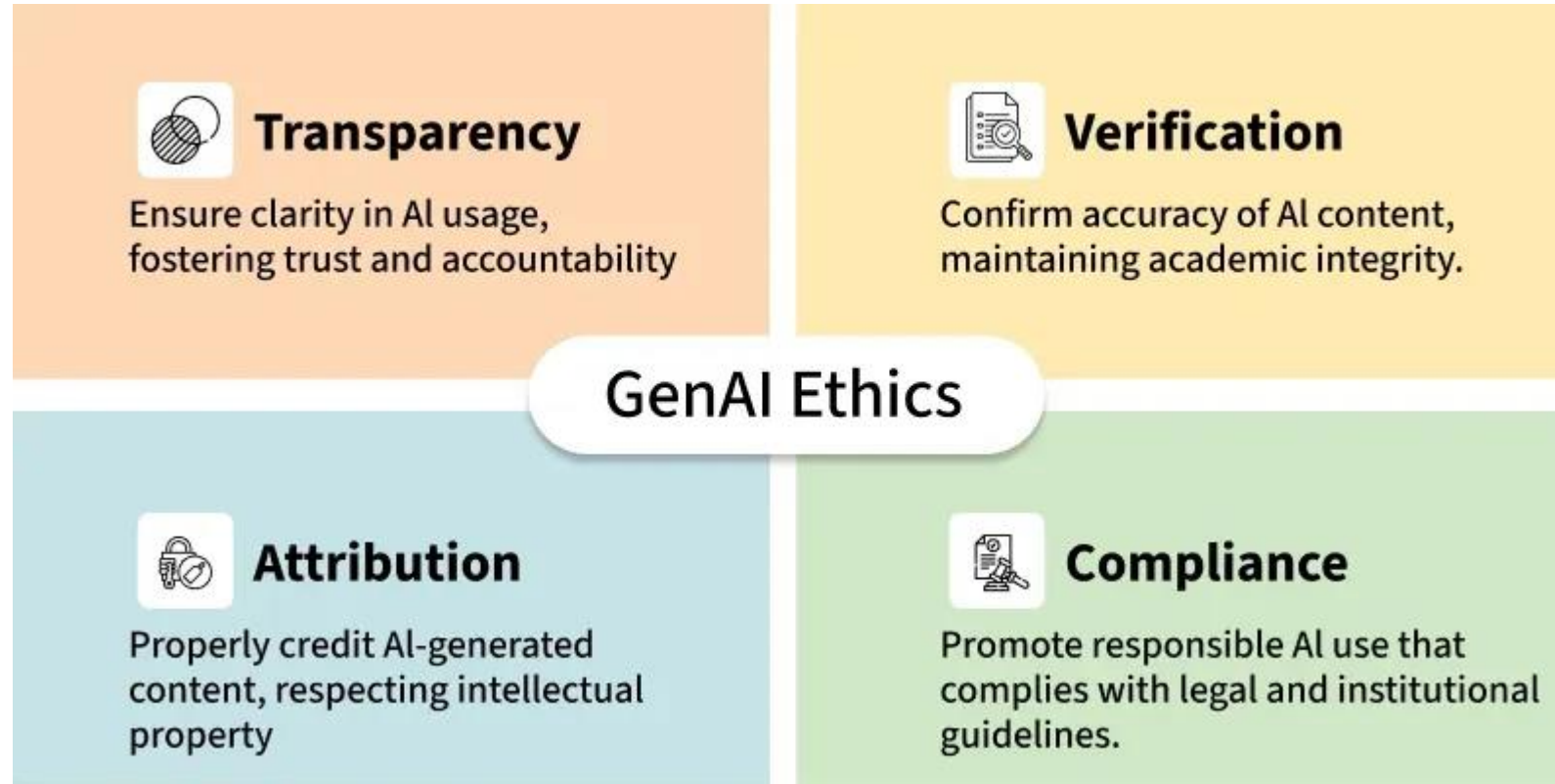


9. AI for Public Good

Responsible AI includes harnessing AI to benefit the public by democratising access, improving public sector adoption, upskilling workers and developing AI systems sustainably

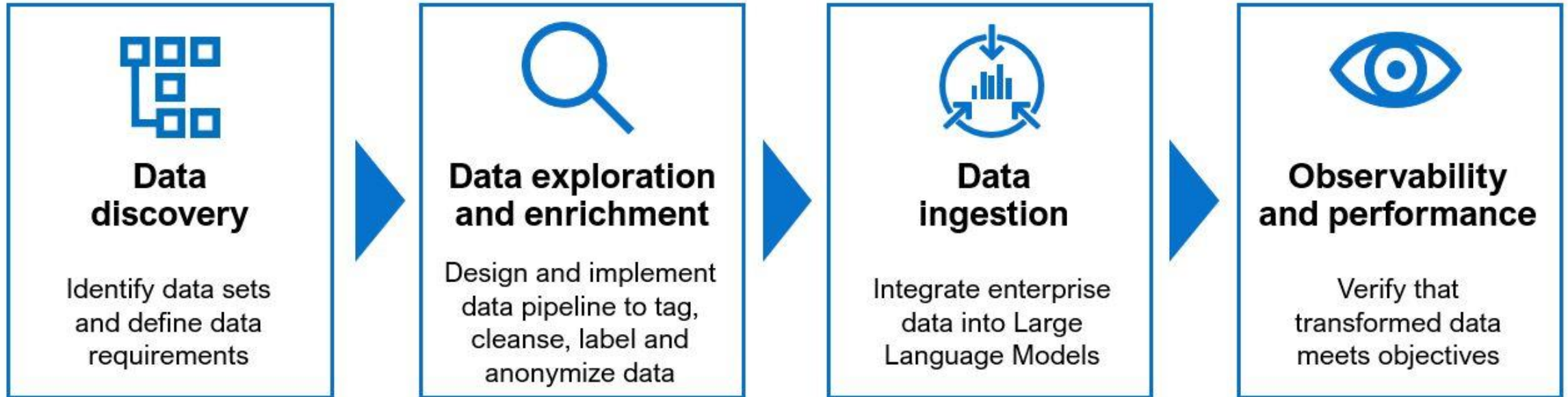


GenAI in Singapore: Accountability



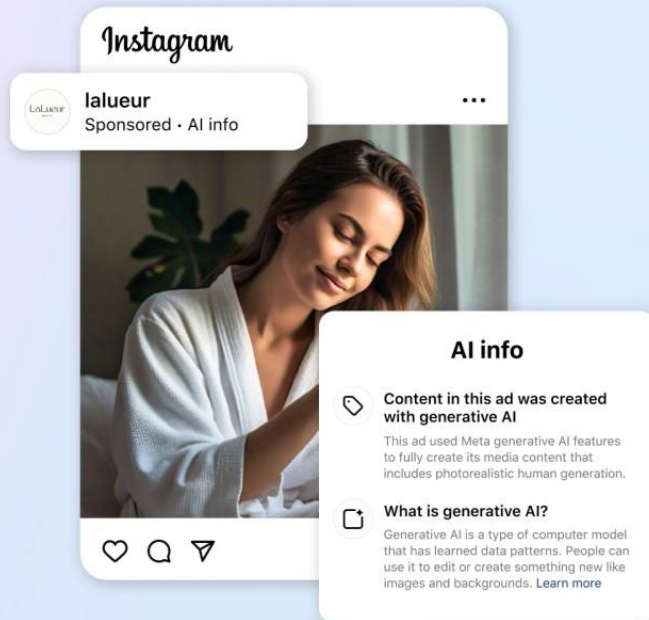
Accountability is a key consideration to incentivise players along the AI development chain to be responsible to end-users. In doing so, we recognise that generative AI, like most software development, involves multiple layers in the tech stack, and hence the allocation of responsibility may not be immediately clear. While generative AI development has unique characteristics, useful parallels can still be drawn with today's cloud and software development stacks, and initial practical steps can be taken.

GenAI in Singapore: Data



Data is a core element of model development. It significantly impacts the quality of the model output. Hence, what is fed to the model is important and there is a need to ensure data quality, such as through the use of trusted data sources. In cases where the use of data for model training is potentially contentious, such as personal data and copyright material, it is also important to give business clarity, ensure fair treatment, and to do so in a pragmatic way

GenAI in Singapore: Trusted Development and Deployment



Model development, and the application deployment on top of it, are at the core of AI-driven innovation. Notwithstanding the limited visibility that end-users may have, meaningful transparency around the baseline safety and hygiene measures undertaken is key. This involves industry adopting best practices in development, evaluation, and thereafter “food label”-type transparency and disclosure. This can enhance broader awareness and safety over time.

Going forward, it is important that the industry coalesces around best practices in development and safety evaluation. Thereafter, meaningful transparency around baseline safety and hygiene measures undertaken will also be key. This will enable safer model use by all stakeholders in the AI ecosystem. Such transparency will need to be balanced with legitimate considerations such as safeguarding business and proprietary information, and not allowing bad actors to game the system.



GenAI in Singapore: Incident Reporting



Even with the most robust development processes and safeguards, no software we use today is completely foolproof. The same applies to AI. Incident reporting is an established practice and allows for timely notification and remediation. Establishing structures and processes to enable incident monitoring and reporting is therefore key. This also supports continuous improvement of AI systems.

Before incidents happen, software product owners adopt vulnerability reporting as part of an overall proactive security approach. They co-opt and support white hats or independent researchers to discover vulnerabilities in their software, sometimes through a curated bug-bounty programme. Once discovered, the vulnerability is reported and the product owner is then given time (typically 90 days, based on industry practice) to patch their software, publish the vulnerability (such as by filing a CVE — see box below) and crediting the white hat or independent researcher. This allows both the software product owners and users to undertake proactive steps to enhance overall security.

GenAI in Singapore: Testing and Assurance

Stress Testing

['stres 'te-stɪŋ]

A computer-simulated technique to analyze how banks and investment portfolios could fare in drastic economic scenarios.

 Investopedia

Third-party testing and assurance often play a complementary role in a trusted ecosystem. We do this today in many domains, such as finance and healthcare, to enable independent verification. While companies typically conduct audits to demonstrate compliance with regulations, more companies are beginning to see external audits as a useful mechanism to provide transparency and build greater credibility and trust with end-users.

While this is an emerging field, we can draw from established audit practices to grow the AI third-party testing ecosystem. Third-party testing will also benefit from comprehensive and consistent standards around AI evaluations (discussed earlier in the Trusted Development and Deployment dimension).



GenAI in Singapore: Security

Abnormal

5 ChatGPT
Jailbreak Prompts
Used by Cybercriminals

Learn More →



Generative AI has brought renewed focus on the security of AI itself. Many issues are familiar, such as supply chain risks in AI/ML middleware. In addressing AI security, it is useful to separate traditional software security concerns addressed via current approaches, from novel threat vectors against the AI model itself. The latter is a nascent space. Nevertheless, similar security concepts may still apply.

Security-by-design is a fundamental security concept. It seeks to minimise system vulnerabilities and reduce the attack surface through designing security into every phase of the systems development life cycle (SDLC). Key SDLC stages include development, evaluation, operations and maintenance.



GenAI in Singapore: Content Provenance



The rise of generative AI, which enables the rapid creation of realistic synthetic content at scale, has made it harder for consumers to distinguish between AI generated and original content. A common manifestation of such concern is deepfakes. This has exacerbated harms like misinformation, and even potential societal threats like undermining the integrity of elections.

There is recognition across governments, industry and society on the need for technical solutions, such as digital watermarking and cryptographic provenance, to catch up with the speed and scale of AI-generated content.³⁵ Digital watermarking and cryptographic provenance both aim to label and provide additional information,³⁶ and are used to flag content created with or modified by AI



GenAI in Singapore: Content Provenance

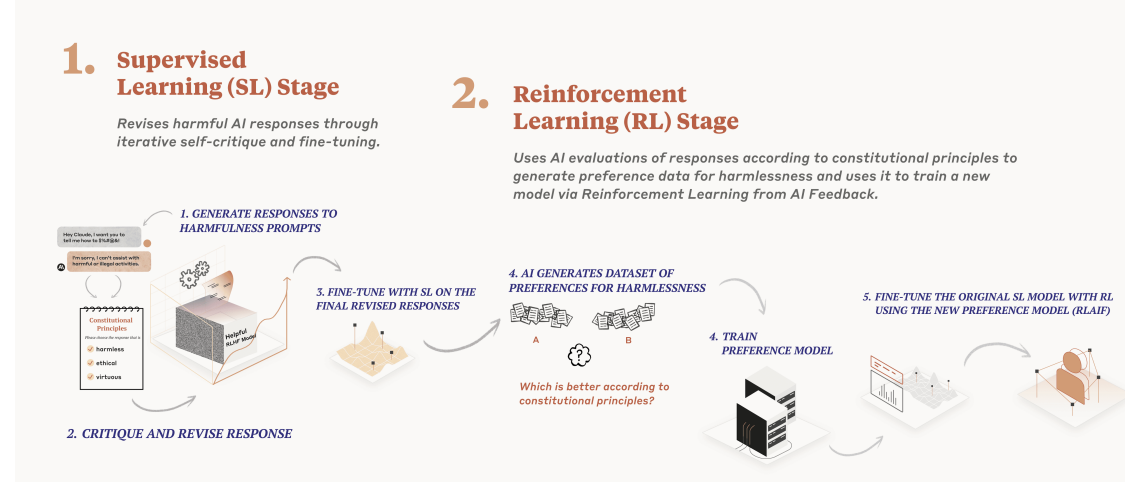


Digital watermarking techniques embed information within the content and can be used to identify AI-generated content. There are several digital watermarking solutions to label AI-generated content today (e.g., Google DeepMind's SynthID and Meta's Stable Signature). However, it is only possible to decode a watermark through the same company that encodes the watermark, due to the current lack of interoperable standards.

Cryptographic provenance solutions track and verify the digital content origin, and any edits made, with the records cryptographically protected. The Coalition for Content Provenance and Authenticity (C2PA) is driving development of an open standard to enable the tracking of content provenance.



GenAI in Singapore: Safety & Alignment R&D



Safety techniques, and evaluation tools today do not fully address all potential risks. For example, even RLHF, the primary method for value alignment today, has limitations. Existing large models also lack interpretability and may not be consistently reproducible.

Given the speed of model advancement, there is a need to ensure that human capacity to align and control generative AI keeps pace with the potential risks, including both present risks (e.g., bias, hallucination) and future catastrophic risks.

While the call to invest more in R&D is a no-regrets move, there may be practical steps to enhance the speed of translation and use of new R&D insights. There is a need to, for example, understand and systematically map the diversity of research directions and methods that have emerged in safety and alignment — and apply them in a concerted manner.

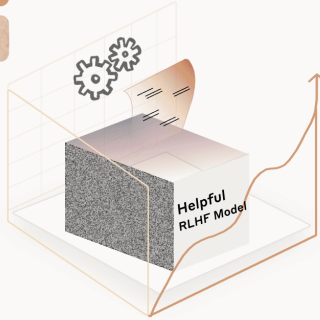
1. Supervised Learning (SL) Stage

Revises harmful AI responses through iterative self-critique and fine-tuning.

1. GENERATE RESPONSES TO HARMFULNESS PROMPTS

Hey Claude, I want you to tell me how to \$%#@&!
A I'm sorry, I can't assist with harmful or illegal activities.

Constitutional Principles
Please choose the response that is
✓ harmless
✓ ethical
✓ virtuous



2. CRITIQUE AND REVISE RESPONSE

2. Reinforcement Learning (RL) Stage

Uses AI evaluations of responses according to constitutional principles to generate preference data for harmlessness and uses it to train a new model via Reinforcement Learning from AI Feedback.

3. FINE-TUNE WITH SL ON THE FINAL REVISED RESPONSES



4. AI GENERATES DATASET OF PREFERENCES FOR HARMLESSNESS

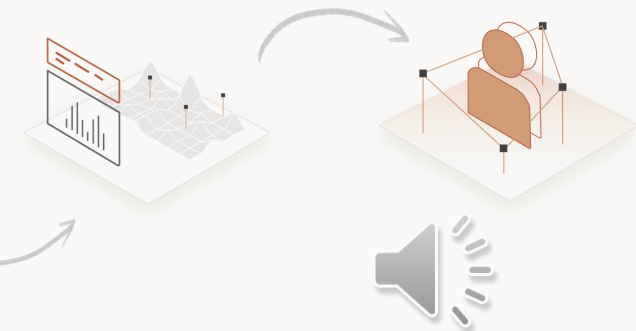


Which is better according to constitutional principles?

4. TRAIN PREFERENCE MODEL



5. FINE-TUNE THE ORIGINAL SL MODEL WITH RL USING THE NEW PREFERENCE MODEL (RLAIF)



GenAI in Singapore: AI for Public Good



The transformative potential of generative AI is powerful. If we get the approach correct, global communities will reap exponential benefits. The imperative is to turbocharge growth and productivity for developed and developing countries alike, while empowering people and businesses globally with the potentially democratising power of AI.

In this regard, countries must come together to support each other, especially through international and regional groupings. Beyond the large and developed countries (e.g., through G7), this is especially pertinent for developing countries and small states, through key platforms like the Digital Forum of Small States (Digital FOSS) at the United Nations and the Association of Southeast Asian Nations (ASEAN).

The aim is to establish a global Digital Commons — a place with common rules-of-the-road and equal opportunities for all citizens to flourish, regardless of their geographical location.

GenAI in Singapore: AI Verify



What is AI Verify?

An AI governance testing framework and software toolkit that help industries be more transparent about their AI to build trust.

[Read more](#)

AI Verify is an AI governance testing framework to help companies assess the responsible implementation of their AI system against 11 internationally recognised AI governance principles.



The Framework

AI Verify testing framework aims to help companies assess their AI systems against 11 internationally-recognised AI governance principles:

- | | |
|------------------------------------|---|
| 1. Transparency | 8. Data Governance |
| 2. Explainability | 9. Accountability |
| 3. Repeatability / Reproducibility | 10. Human Agency and Oversight |
| 4. Safety | 11. Inclusive Growth, Societal and Environmental Well-being |
| 5. Security | |
| 6. Robustness | |
| 7. Fairness | |

AI Verify testing framework is consistent with other international AI governance frameworks such as those from ASEAN, European Union, OECD and the US.

WHO SHOULD USE THE FRAMEWORK?

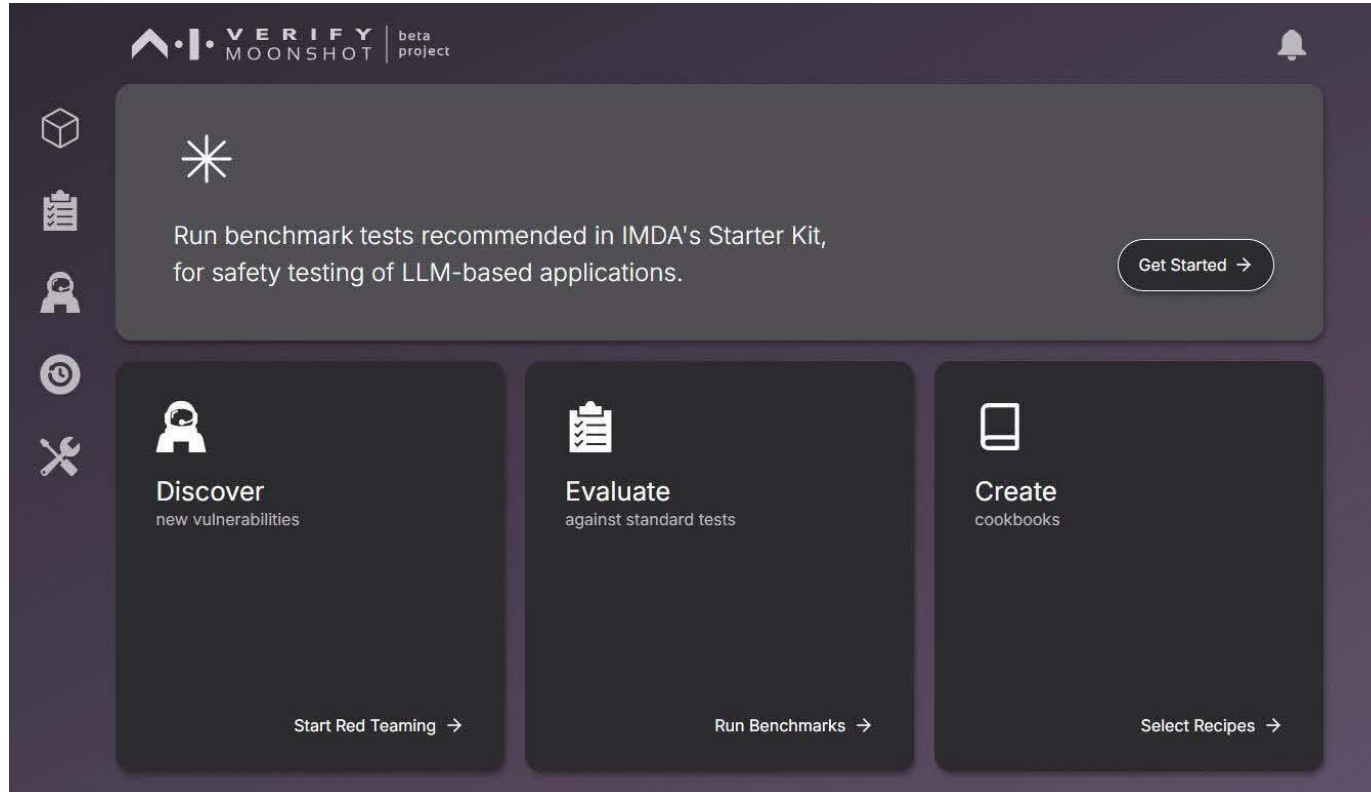
AI System Owners /
Developers looking to
demonstrate their implemen
tation of responsible AI
governance practices

Internal Compliance
Teams looking to ensure
responsible AI practices have
been implemented

External Auditors looking to
validate your clients'
implementation of
responsible AI practices



GenAI in Singapore: Project Moonshot



Project Moonshot is one of the world's first Large Language Model (LLM) Evaluation Toolkits. It is an open-source tool to bring benchmarking and red teaming together to help LLM App developers and compliance teams to test and evaluate their LLMs and LLM applications.



The End Questions?

